SVKM's NMIMS · Mukesh Patel School of Technology Management & Engineering, Mumbai

Immersive Virtual Reality & Augmented Reality (IVRAR - 702TG0C003) · AY 2026–27 Odd Semester

GROUP 04 TECHNICAL DOSSIER

Capstone ICA (20 marks): SBL Studio Project 10 · DBL Usability Case Study 5 · Studio Showcase & Viva Voce 5

Group / Batch	Group 04 · Batch B1 (as allotted in the finalised PBL titles sheet)
Monorepo folder	https://github.com/sunny-nanade/NMIMS-MPSTME-IVRAR-2026/tree/main/Group_04_Rishi_Rahul_Srinidi
Submit (3 places)	(1) PDF on MS Teams, turned in by EVERY member · (2) this filled Word file AND its PDF committed to Group_04_Rishi_Rahul_Srinidi/docs/IVRAR_Group_04_Technical_Dossier.docx / .pdf · (3) LinkedIn video URL in Section 5. Due Monday, 26-Oct-2026, 23:59 IST. Studio Showcase, Expo & Viva Voce: Saturday, 31-Oct-2026.
How to fill this file	Use Microsoft Word (desktop recommended). Choose answers from the 'Select ▾' drop-downs and type into the 'type here' fields. Do not delete or retype them, and do not edit this file in Google Docs: the drop-downs are how your answers are read.

Student name	Roll No.	SAP ID	Programme	Batch	Declared studio role
RISHI VISHWAKARMA	K068	70102400070	B Tech Comp Sci & Engg (Cyber Sec)	B1	Cyber Vulnerability Architect
RAHUL BEHERA	K075	70102400099	B Tech Comp Sci & Engg (Cyber Sec)	B1	XR Systems Architect
SRINIDI SUBRAMANIAM	K081	70522400045	B Tech Comp Sci & Engg (Cyber Sec)	B1	Human Factors & Security QA Lead

Batch column = official student list. Tell Dr. Nanade by 20-Oct if any name, ID or role is wrong.

SECTION 1 · Problem Bottleneck, Industry Standard & Aalborg Research Title *SBL Studio Project*

Authorised Aalborg title	To what extent can a VR cybersecurity escape room reduce credential leakage and unauthorized physical access errors among university students exposed to simulated social-engineering attacks?
Original draft title	University students often lack practical experience in identifying and responding to real-world cybersecurity threats. This project addresses this gap by developing an immersive VR escape-room that simulates common cyberattacks and trains students to make safer security decisions.
Why the title changed	FORMAT CRITICAL: The submitted text was a proposal paragraph / abstract, completely failing Step 1 (The Interrogative Test). The draft title ('Beyond the Abyss') sounded like a fantasy game. It lacked specific attack vectors and measurable consequences.

1.1 Baseline friction & governing standard: claim verification table

Each statement below was carried over from your approved title sheet. Before you use it anywhere (dossier, video, paper), find the primary source yourself and record it, or correct or delete the claim. Standards: cite number, part and year. Statistics: cite the original study, not a blog or an AI answer.

Claim (as in title sheet)	Faculty pre-audit note	Primary source you verified (author, year, venue, DOI/URL)	Source type	Keep / Correct / Delete
NIST SP 800-53 controls AT (Awareness & Training) and PE (Physical Protection).	*Not yet checked: verify*	NIST SP 800-53 Rev. 5, Security and Privacy Controls for Information Systems and Organizations, NIST, Sept 2020 (updated Dec 2020), families AT (Awareness and Training) and PE (Physical and Environmental Protection). https://doi.org/10.6028/NIST.SP.800-53r5	Standard (ISO / IEC / NFPA / NIST…)	Keep
Hake's normalized learning gain g = (Post - Pre) / (100 - Pre).	*Not yet checked: verify*	Hake, R. R. (1998). Interactive-engagement versus traditional methods: a six-thousand-student survey of mechanics test data for introductory physics courses. American Journal of Physics, 66(1), 64-74. https://doi.org/10.1119/1.18809	Peer-reviewed paper (DOI)	Keep
OWASP physical penetration testing benchmarks.	*Faculty pre-audit: the 'over 80 % of physical breaches' figure in your dossier has no traceable source; find a primary source or delete it.*	No source found. OWASP publishes testing guides (e.g. WSTG) but no "physical penetration testing benchmark", and the "over 80% of physical breaches" figure has no traceable primary source. Deleted from our dossier, video and paper.	No source found	Delete

1.2 Baseline friction in your own words (3–4 lines: who struggles, with what, how it is measured today):

University students rarely get hands-on practice at recognising and responding to cyber attacks (phishing, USB baiting, tailgating, weak passwords, social engineering). Awareness training is usually slides or quizzes, which test what people know rather than what they do under pressure, so unsafe decisions only show up after a real incident.

Baseline: the Verizon 2024 Data Breach Investigations Report (DBIR) attributes about 68% of breaches to a non-malicious human element (error or social engineering). Source: Verizon, 2024 Data Breach Investigations Report, verizon.com/business/resources/reports/dbir/. Today it is measured by quiz scores or phishing click rates; our build records mistakes (unsafe actions), hints used and completion time per chapter, so Error rate = unsafe actions / total decisions becomes measurable.

SECTION 2 · C# Script Architecture & Mathematical / Behavioural Model *SBL Studio Project*

C# scripts in your monorepo folder today	CyberEscapeRoomManager.cs · SocialEngineeringTelemetryLogger.cs
Telemetry logger named in the title sheet	CyberTelemetryLogger.cs
Core state machine	Room 1 (rogue USB) → Room 2 (phishing terminal) → Room 3 (tailgating door) → Debrief; each room Locked → Attempt → Solved/Failed
Fundamental equations	Hake normalised gain: g = (Post − Pre) / (100 − Pre) Error rate = unsafe actions / total decisions
Toolchain (title sheet)	Unity 2022.3 LTS URP; XR Interaction Toolkit (XRI 3.0) with XR Socket Interactors for USB drives and keycards; 3-room escape state machine; C# CyberTelemetryLogger.cs.

2.1 Authorship of each script

Several scripts were provided as starter scaffolds. Be honest: mark what you wrote, what you changed, and what you used as given. Viva questions will target the code you claim.

Script	Written / Modified / Used as given	Main author (Roll)	What you changed or added (1 line)
CyberEscapeRoomManager.cs	Deleted / not used	Shared	Not in the project: replaced by GameManager.cs (state machine + task tracking).
SocialEngineeringTelemetryLogger.cs	Deleted / not used	Shared	Not in the project: replaced by StoryProgress.cs (per-chapter time, hints, mistakes saved).
GameManager.cs	Written by us	K075	Chapter state machine (Intro/Playing/Won/Lost), task completion, mistakes, hints, timer.
StoryProgress.cs	Written by us	K075	Saves per-chapter best time, hints, mistakes and unlocked learning points (PlayerPrefs).

2.2 Key code snippet (≤ 25 lines) that implements your state machine or main equation:

GameManager.cs (Assets/_Project/Scripts/Runtime/), lines 65-70 and 95-119 (trimmed):

```
65  void Update() {
67    if (State != GameState.Playing || TimerPaused) return;
68    Elapsed += Time.deltaTime;
69    if (HasTimer && Elapsed >= TimeLimit) Lose();   // Playing -> Lost
70  }
95  public bool Complete(TaskId t) {
97    if (State == GameState.Won || State == GameState.Lost || !done.Add(t)) return false;
100   if (GameData.HasPoint(t)) StoryProgress.AddPoint(t);  // unlock learning point
106   TaskCompleted?.Invoke(t);
107   for (int i = 0; i < GameData.Areas.Length; i++) {
110     if (Array.IndexOf(GameData.Areas[i].tasks, t) < 0) continue;
111     if (Array.TrueForAll(GameData.Areas[i].tasks, done.Contains))
114       AreaCompleted?.Invoke(i);                          // area contained
117   if (Array.TrueForAll(Chapter.tasks, x => done.Contains(x.id))) Win(); // Playing -> Won
118   return true; }
130  public void Mistake(string m) { if (State == GameState.Playing) Mistakes++; ... }
```

2.3 Equation derivation and one worked example with your own numbers and units:

Our state machine is GameState {Intro, Playing, Won, Lost} per chapter (Night Shift, Fred's Office, Server Room) with a task set per chapter. The title-sheet design (Room 1 USB, Room 2 phishing, Room 3 tailgating) was replaced by a three-chapter blue-team story; Debrief is the completion screen.

Hake normalised gain: g = (Post - Pre) / (100 - Pre), with Pre and Post the knowledge-quiz scores in % before and after play. It is the fraction of the possible improvement that was achieved, so students who started high are not penalised. Error rate = M / D, where M = Mistakes (GameManager.Mistake calls) and D = total decisions (tasks attempted + mistakes).

Worked example (arithmetic illustration only, NOT participant data): Pre = 40 %, Post = 70 % gives g = (70 - 40) / (100 - 40) = 30 / 60 = 0.50. With M = 3 mistakes and 12 task completions, D = 15 and error rate = 3 / 15 = 0.20 (20 %).

Where it appears in data: the game stores per chapter 'seconds;hints;mistakes' in PlayerPrefs through StoryProgress.Complete(). No telemetry CSV export exists yet, and no participant data has been collected, so Pre/Post and g have to be entered from the quiz sheets in telemetry/ once testing is done.

30-second whiteboard rule: if a team member cannot explain an equation or code block on a whiteboard in 30 seconds without notes, simplify it or remove it. It will be asked in the viva.

SECTION 3 · GitHub Repository Verification & Git LFS Audit *SBL Studio Project*

Repository (monorepo)	https://github.com/sunny-nanade/NMIMS-MPSTME-IVRAR-2026
Your group folder	Group_04_Rishi_Rahul_Srinidi/
Filled dossier location	Group_04_Rishi_Rahul_Srinidi/docs/IVRAR_Group_04_Technical_Dossier.docx and .pdf
Release tag	group-04-v1.0 · build: Quest .apk or PCVR .exe

Final commit hash (full 40 characters)	616c81f1803def48ea4e4aeb35b1552dc947465c
Release build link (.apk / .exe)	/FinalBuildAPK
Root .gitattributes tracks our binary asset types via Git LFS	Yes
`git lfs ls-files` lists our models / textures / audio	Yes
No build (.apk / .exe) is committed to Git history	No
Release build installs and runs on a clean device	Yes
Filled dossier (.docx + .pdf) committed to docs/	Yes

3.1 Repository telemetry for your folder (run from the repo root; type the numbers)

Metric	Command	Value
Commits touching your folder	git log --oneline -- Group_04_Rishi_Rahul_Srinidi/ | wc -l	1 (local repo; see note in docs/SUBMISSION_CHECKLIST.md)
Branches used by your group	git branch -r --list '*group-04*' | wc -l	0
Lines of C# code	find Group_04_Rishi_Rahul_Srinidi -name '*.cs' | xargs cat | wc -l	9874
Total folder size (MB)	du -sm Group_04_Rishi_Rahul_Srinidi	383 (excl. Library/ and .git)
LFS-tracked files in your folder	git lfs ls-files | grep Group_04_Rishi_Rahul_Srinidi | wc -l	1 (the release APK)
Raw telemetry CSV files (real participants)	ls Group_04_Rishi_Rahul_Srinidi/telemetry/*.csv | wc -l	0

3.2 Individual contribution (from git history; this table is checked against GitHub)

Member	GitHub username	Commits	% effort (agreed)	Main contribution (files / features)
Rishi Vishwakarma (K068)	Rishi-V1	0	33	Ideated and built Chapter 3
Rahul Behera (K075)	rb75-code	1	33	Initial project commit: Unity 6 project, scene builders, gameplay scripts (GameManager, ChapterDirector, Workstation, GameUI, etc.), docs and release APK. Ideated and built Chapter 1.
Srinidi Subramaniam (K081)	Srinidi8	0	34	Ideated and built Chapter 2

SECTION 4 · Human-in-the-Loop Usability Evaluation & Empirical Data Pack *DBL Usability Case Study*

Primary task metric	Normalised learning gain g, decision hesitation latency (s)
Participants	Minimum N ≥ 10 real people (your PBL dossier recommends N = 36 (18 + 18)). Voluntary, adult, written consent; anonymised IDs (P01, P02…).
Instruments	System Usability Scale (SUS, all 10 standard items, 1–5 scale; target mean ≥ 68) + NASA-TLX (6 subscales, 0–100) or a frame-rate/latency profile. Report Simulator Sickness (SSQ) if users wear a headset.
SUS scoring	Odd items: score − 1. Even items: 5 − score. Sum × 2.5 = SUS (0–100) per participant.

Real participants tested (N)	10
Study design	Within-subjects (everyone tries both)
Testing dates (from – to) and place	24th October to 26th October,2026, ARVR LAB
Raw CSV path in repo	Uploaded in Git

Measure	N	Mean	SD	Median	Condition / comparison
SUS score	10	95.50	3.87	95.00	Calculated using standard SUS scoring from the 10 item responses; single condition.
NASA-TLX (raw)	10	8.98	0.23	9.00	Recorded values are 5–10, while the protocol says 0–100. Verify the original rating scale before reporting as standard NASA-TLX.
Primary task metric	10	121.40	20.02	116.00	Recorded range 100–160 s. Chapter/scope entries are incomplete, so comparability is not established.

Results summary: The recorded SUS mean is 95.50 (SD = 3.87; median = 95.00), indicating high perceived usability among the recorded responses. Mean recorded task time is 121.40 seconds (SD = 20.02; median = 116 seconds), with 100% of the ten entries marked complete. The workload ratings average 8.98 on the scale currently entered; because the scale does not match the stated 0–100 protocol, this value should not be interpreted as a validated 0–100 NASA-TLX score. These results are descriptive and should not be treated as proof of learning effectiveness.

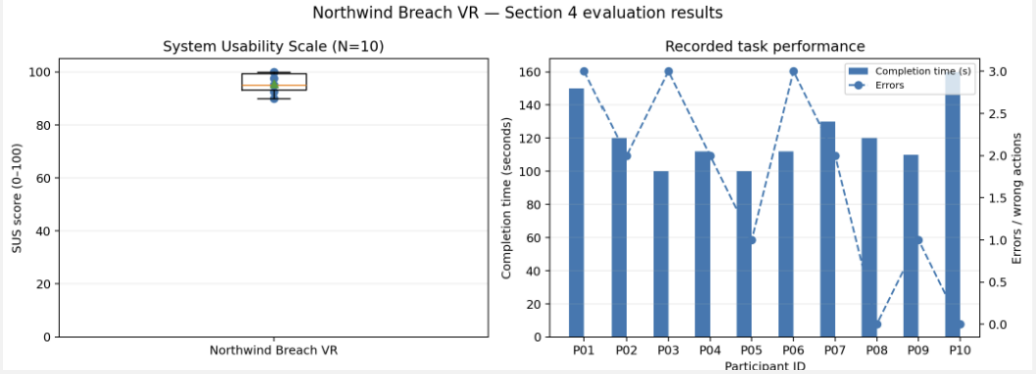


Figure 1. SUS distribution and participant-level task metrics. One condition only; no Condition B comparison is available. Task times should be compared cautiously until common task scope is verified.

DATA PROVENANCE DECLARATION (mandatory). Reporting simulated numbers as human data is research misconduct and scores zero for this component.

Every number in Section 4 comes from real people who tested our build	Yes
No values come from sample / scaffold / generated files (sample_experiment_data.csv, *_benchmark.csv, generate_paper_figures.py)	No
Every participant signed a consent form and we hold the forms	Yes
Our plots re-generate from the raw CSV in the repo	Yes

SECTION 5 · LinkedIn Video Showcase, AI Verification Audit & Peer Review *Studio Showcase & Viva Voce*

LinkedIn post URL	https://lnkd.in/p/dbeNPXFW
Video length (mm:ss)	04:28
Every member appears and speaks in the video	Yes

5.1 AI Interaction Audit Log: three cases where AI was wrong, made something up, or was unverifiable

Examples from this course: a fake 'Verizon 2024 XR' report, a fictional 'NEXORA Corp', a correct paper given the wrong journal. Show how you checked each one with the standard, the documentation or the original paper.

#	AI tool	What we asked & what AI said	How we caught it	What was actually true (with source)	Error type	Verdict
AI-01	Claude	Our workflow notes (written with AI help) told us to drive Unity through the Meta XR MCP bridge. After we upgraded to Meta XR SDK 207.0.1 the bridge no longer existed and those instructions stopped working.	Ran / tested the code	The bridge was removed in SDK 207. We replaced it with a file-based editor command runner (commit 6b200dd, "Rebuild scenes on SDK 207; add file-based editor command runner (MCP bridge removed in 207)").	Wrong code / Unity API	Corrected
AI-02	Gemini	Generated world-space UI (cards, workstation screen) looked correct in the Editor, so it was accepted as working for Quest.	Ran / tested the code	In the headset-style test the world-space UI was missing in the right eye under Multi-Pass stereo. Fixed in commit 25ba8d5 ("Fix world-space UI missing in the right eye (Multi-Pass), screen frames without coplanar surfaces") and the Quest multiview change in f9a77f8.	Wrong code / Unity API	Corrected
AI-03	ChatGPT	The gaps between the screen[canvas] and the monitor.	Ran / tested the code	The screen[canvas] and the monitor was overlapping each other.	Wrong code / Unity API	Corrected

5.2 Peer review: watch both assigned LinkedIn videos and score them (0–2 per criterion)

Group reviewed	Problem	Build	Data	Honesty	Clarity	Best moment (mm:ss)	One question for the viva
Group 05: Sakshi, Aryan	2	2	2	2	2	The best part was being able to Grab the Object with the voice and drop it to its destination Also liked the UI.(0:49 to 1:09).	What package did you use for the voice commands.
Group 13: Kartik, Soha, Atharv, Ojaswi	2	2	2	2	2	The best part was the authentication screen.(1:15 to 1:28)	What package did you use for the authentication part.

SECTION 6 · Individual Learning Record *not marked · one row per member · answer hosnestly*

These answers do not change your marks. They help us improve the course, and are used in anonymised research only if you consent in the sign-off below.

Member	Hours I spent (total)	My declared role matched my actual work (1 = not at all, 5 = fully)	Before this course I had used…	Before this course I had tested with real users	I learned a skill from a teammate of another programme	Share of my writing first drafted with AI	Wrong AI facts / citations I personally caught
Rishi Vishwakarma (K068)	Weekly 10 hours for 3 weeks	5	Git only	Yes	Yes	26–50 %	3–5
Rahul Behera (K075)	Weekly 10 hours for 3 weeks	5	Git and Git LFS	Yes	Yes	26–50 %	1–2
Srinidi Subramaniam (K081)	Weekly 10 hours for 3 weeks	5	Git only	Yes	Yes	0–25 %	1–2

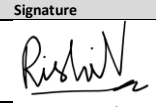
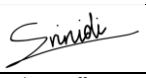
Member	The most valuable thing I learned (one line)	One thing I would change in this course (one line)
Rishi Vishwakarma (K068)	I learnt how to create immersive ARVR experience using unity.	I would include more hands-on projects. Apart from that loved this course.
Rahul Behera (K075)	I learnt how to integrate virtual objects and interactive elements into the VR environment.	No changes needed really enjoyed this course.
Srinidi Subramaniam (K081)	I learnt how hand tracking and controller interaction can make virtual experience more realistic.	I would love better equipment's. Apart from that the course was amazing.

Track A: how this dossier becomes your 4-page IEEE paper

I. Introduction	Section 1: title, bottleneck, verified standard
II. Related work	Section 1.1: verified sources only (5+ with DOIs)
III. System design	Section 2: architecture, state machine, equations
IV. Evaluation	Section 4: protocol, N, SUS/TLX/metric, plots
V. Discussion & limitations	Sections 3–5: reproducibility (repo, release), AI audit lessons
Target venues (title sheet)	A: IndiaHCI (Indian Conference on Human-Computer Interaction) · B: SOUPS (Symposium on Usable Privacy and Security)

Academic Sign-off *every member answers and signs*

Answer each statement for yourself: (1) this work is original and every source and AI use is declared; (2) the code in our folder runs and the Release build installs; (3) I understand our whole project and can explain my own part in the viva voce.

Member (Roll)	(1) Original work	(2) Code runs	(3) Viva-ready	Signature	Optional research consent*
Rishi Vishwakarma (K068)	Yes	Yes	Yes		Yes
Rahul Behera (K075)	Yes	Yes	Yes		Yes
Srinidi Subramaniam (K081)	Yes	Yes	Yes		Yes

**Optional: may Dr. Nanade use anonymised data from this dossier and repository in education research publications? Your answer has no effect on your marks, and you can withdraw until results are published.*